

Mode of Examination: Online
M Sc. Semester – II Examination, 2021
Subject: Computer Science

Paper Code & Name: CSM 202 Design and Analysis of Algorithms

Date: 11.08.2021

Full Marks: 70

Time and Duration: 12 pm to 3.00 pm

Please note the following instructions carefully:

Promise not to commit any academic dishonesty.

Answer in your own words as far as applicable.

Each page of the answer scripts should have your University Roll # on the right-top corner.

The name of the scanned copy of the answer script will be of the following format:

Paper code-Paper Name-My Roll number.pdf

(Example: CSM-202-DAA-C91-CSC-201001.pdf)

The subject of the mail should be the file name only.

The scanned answer-script is to be sent to **cucse2020@gmail.com**

The report should have an index page.

30 minutes is allotted for scanning and uploading your answer script.

The answer-script may not be accepted after the scheduled time.

Answer questions 1,2 and any four from the rest.

1. Answer any five questions

$5 \times 2 = 10$

- (a) Explain why quicksort algorithm is not stable.
- (b) Given functions $f(n)$ and $g(n)$ such that $0 < f(n) < g(n)$ for all n , and a constant $0 < \alpha < 1$, and a function $h(n) = \alpha f(n) + (1 - \alpha)g(n)$, show that $h(n) = \Omega(f(n))$ and $h(n) = O(g(n))$
- (c) Let $f(x) = x^2 + (n - x)^2$. Find $\max_{x=0,1,\dots,n} f(x)$.
- (d) Explain why absence of negative weight cycles is necessary for the shortest path problem.
- (e) Mention the properties a graph should satisfy so that a minimum spanning tree for the graph can be computed.
- (f) Is $2^{2n} = O(2^n)$? Justify.

2. Answer any 5 questions

5 × 4 = 20

- (a) Find $T(n)$ in terms of big-oh and big-omega notation when
 $T(n) = T(\lfloor n/2 \rfloor) + T(\lceil n/3 \rceil) + \sqrt{n}$. Assume $T(1) = 1$ and $T(2) = 2$
- (b) There are n points in a plane $p_1, p_2, \dots, p_n$. Let $p_i.x$ and $p_i.y$ denote the x and y co-ordinates of p_i . Suppose you have to set up a facility f at some location $(f.x, f.y)$. Cost of the facility is equal to $\sum_{i=1}^n (d(f, p_i))^2$ where $d(f, p_i)$ is the Euclidean distance between f and p_i . Give an algorithm to find the location of the facility such that the cost is minimum.
- (c) You have to sort 2^{24} 48-bit numbers using radix sort. Suppose you group the 48 bits into groups of k bits and use as radix 2^k for each pass. Find the value of k which will minimize the execution time of the radix sort.
- (d) Suppose you have an array $A[1..2n]$ where the subarray $A[1..n]$ and $A[n+1..2n]$ are sorted in ascending order. Let M be the median of the elements of A . Let $p = \lfloor \frac{n+1}{2} \rfloor$.
Prove that if $A[p] < A[n+p]$, the median M is the median of the subarray $A[p..n+p]$
- (e) Explain the significance of polynomial time reducibility for developing the concept of NP completeness.
- (f) Write an algorithm to find the k^{th} largest element of an array of $A[1..n]$ which takes time $O(n)$ in the worst case.
3. Let B be the total number of bit set/reset operations performed for incrementing from 0 to a k -bit integer $n = a_{k-1}a_{k-2} \dots a_1a_0$, where $a_{k-1} = 1$, and its binary representation has p 1's and $k - p$ zeroes. Find B in terms of n , p and k .

Show that the cost of n insertions in a dynamic table with load factor $\alpha > 1/k$ where k is a positive integer greater than 2, is $2n + n/(k - 1)$. Explain using accounting method how the amortized cost for each operation is $2 + 1/(k - 1)$. 4+3+3

4. Consider the problem

Given $0 < c_1 \leq c_2 \leq \dots \leq c_m$, and $\sum_{i=1}^m i.x_i = n$, where $n > m$ and each x_i is either zero or positive integers, maximize $\sum_{i=1}^m c_i x_i$.

Give an $O(mn)$ algorithm to solve the problem.

Apply your algorithm to the case where $m = 7$, $c_1 = 2$, $c_2 = 5$, $c_3 = 8$, $c_4 = 9$, $c_5 = 12$, $c_6 = 17$, $c_7 = 18$ and $n = 10$ 6+4

5. You have to find one good chip among n chips. Let G and B be the number of good chips and bad chips such that $G > B$. The chips are tested by putting two of them in a circuit. A good chip will correctly indicate whether the other chip is good or bad. But the diagnosis of a bad chip may or may not be correct.

Show that when n is even, by pairwise checking we can reduce the set to at most half its size maintaining the property that $G > B$ within the reduced set. 2

Let g be the number of pairs which are indicated to be both good (which imply both chips are good or both chips are bad) by tests.

Because we are checking them in pairs one chip remain untested when n is odd.

Show that the invariant $G > B$ is maintained in the reduced set if

i) when g is even, by selecting one from each of the g pairs and including the untested chip. **3**

ii) when g is odd, by selecting one from each of the g pairs and excluding the untested chip. **3**

What is the time complexity of the algorithm which recursively reduces the set until it is of size 1? **2**

6. Consider a connected undirected unweighted Graph $G(V, E)$. Another graph $H(V', E')$ is constructed as follows.

i) $V' = V$

ii) $(u, v) \in E'$ if and only if $(u, v) \notin E$

(a) Prove that if $W \subseteq V'$ is a vertex cover in H then $V - W$ is a clique in G **5**

(b) Prove that if $W \subseteq V$ is a clique in G then $V' - W$ is a vertex cover in H **5**

7. Show that the expected cost of quicksort to sort an array of n elements is $T(n)$ where

for $(n > 0)$, $T(n) = n + \frac{2}{n} \sum_{i=0}^{n-1} T(i)$, and $T(0) = 0$.

Then solve the recurrence to show that for $n > 0$ $T(n) \leq 1.387(n + 1) \log_2 n - (n - 2)$ **3+7**

8. Write Bellman-Ford algorithm for single source shortest path. Apply your algorithm to find the shortest paths from x to all other vertices for the graph shown below. Show the distance estimation after every iteration. **5+5**



